

LOHIT

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SUMMARY

Backend engineer who builds for failure and proves recovery instead of claiming it. Built dex, a Go durable-execution engine whose crash recovery is verified by a deterministic simulator that reproduces entire cluster runs from a single seed, and Forge, an AI incident-investigation backend running a REST API, three async workers and pgvector search inside one free-tier process. B.Tech 2026.

EXPERIENCE

Smartworks — Gurugram, India

Software Development Intern | Jun 2025 – Sep 2025

- Deployed and customized Twenty (open-source CRM) from a local evaluation setup into a self-hosted, team-facing internal product.
- Diagnosed KPI-counting and ticket-assignment defects in the internal ticketing service by tracing incorrect results back through MongoDB aggregation pipelines to the failing stage.
- Built Elasticsearch-backed search across snag and user records, replacing direct collection scans.
- Implemented FCM push notifications for WorkCtrl, a React Native app live on the App Store and Play Store, and integrated Formbricks feedback capture across the app and marketing site.

PROJECTS

dex — Durable Execution Engine | [GitHub](#)

Go, PostgreSQL, deterministic simulation testing

- Designed a single-goroutine deterministic simulator driving every node, the clock and the scheduler from one seeded PRNG, making whole-cluster runs reproducible from a seed and enabling automatic shrinking of failing runs — one triage reduced 44 injected faults to the 2 that caused the bug.
- Engineered a Temporal-style durable execution engine in which workflows resume at the exact step they stopped after a process crash, reconstructing state by replaying an append-only Postgres event log rather than persisting mutable runtime state.
- Implemented shard leases with fencing tokens to prevent split-brain writes, transfer and timer queues with ack levels, and a determinism linter that rejects non-replayable calls (e.g. time.Now) inside workflow code.
- Verified crash recovery by killing a live process mid-execution against Postgres; shipped dexctl (CLI) and an embedded dashboard for inspecting event histories and live workflow state.

Forge — AI Incident-Investigation Backend | [Live Demo](#) | [GitHub](#)

Node.js, Fastify 5, PostgreSQL/pgvector, Redis/BullMQ, Groq, Gemini, Railway

- Sole developer of a backend that ingests alerts from Grafana, PagerDuty, Sentry, CloudWatch and Opsgenie and runs autonomous LLM root-cause investigations, embedding each incident into pgvector so prior failures inform new ones.
- Hardened ingestion against duplicate and racing webhooks using incident fingerprinting and atomic Postgres upserts, with field redaction, at-rest encryption and RBAC-scoped multi-tenant isolation.
- Built a provider-agnostic LLM seam (Groq for generation, Gemini for embeddings) with two-tier model routing and quota-aware rate limiting, so one provider's quota exhaustion degrades the pipeline instead of breaking it.
- Architected the REST API and three BullMQ/Redis background workers to run in a single Fastify 5 process over Postgres.

Real-Time Results Platform | [GitHub](#)

Node.js, Express.js, SQLite, Socket.io, JWT

- Shipped a public results site with a JWT-authenticated admin panel, broadcasting each published result to all connected clients over Socket.io rather than polling, behind a rate-limited REST API.

EDUCATION

Guru Jambheshwar University of Science & Technology — Hisar, India

B.Tech, Electronics & Communication Engineering | Graduated May 2026

Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS, Object-Oriented Programming, System Design

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, Go, Python, SQL, Java, C/C++

Backend: Node.js, Fastify, Express.js, REST APIs, Socket.io, BullMQ, background workers, event sourcing, rate limiting, JWT auth, RBAC, multi-tenant isolation

Databases: PostgreSQL, pgvector, MongoDB, Redis, SQLite, Elasticsearch

Infrastructure: Docker, GitHub Actions, CI/CD, Linux, Railway, Vercel

AI: LLM APIs (Groq, Gemini), embeddings, vector search, prompt engineering

Frontend: React, Next.js, React Native, Tailwind CSS, Drizzle ORM

ADDITIONAL

- Smart India Hackathon college-level qualifier — led development of an alumni-association platform for a government engineering college.
- Maintained the official Training & Placement Cell website, Guru Jambheshwar University.